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Multi-Client Delegated Verifiable Blind Quantum Computing

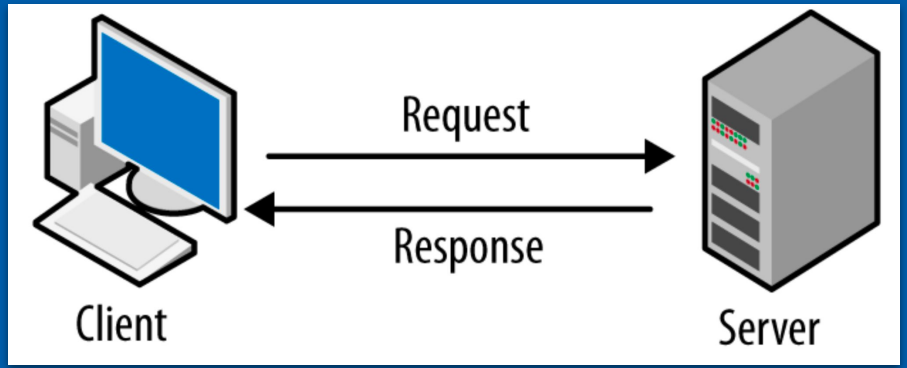
ACM PQQS 2026

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VBDQC

A Client/Verifier wishes to <u>Delegate</u> a quantum circuit to a remote Server/Prover:

- The client may wish to **<u>Verify</u>** that the received response is correct
- The client may wish to keep the server **<u>Blind</u>** to their private input data
- The client may possess some quantum capability or none at all
- Multiple clients may wish to delegate a circuit on their joint **<u>Private</u>** inputs

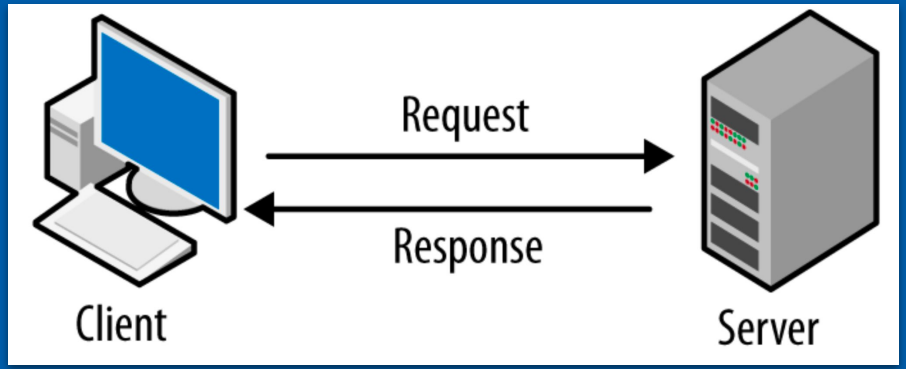


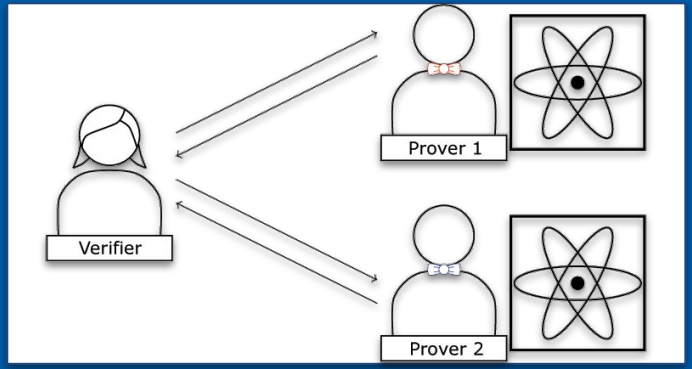
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Existing Solutions

Overview [GKK18]:

- Prepare-and-send [FK17], receive-and-measure [MF16] protocols
  - Typically require weak client quantum capability
- Computational security protocols [Mah18]
  - One server, **classical client**, uses quantum resistant cryptographic assumptions (LWE)
  - Requirements $O(g^4)$ [GV19]
- Entanglement-based protocols [RUV12]
  - Purely **classical client**, multiple non-communicating entangled servers
  - Requirements $O(g \log g)$ [CGJV24]





Multiparty Challenge

MPC oracles:

- Classical MPC protocols (can be made quantum resistant) that construct an ideal single client

Can we just replace the single client with an MPC oracle?:

- Collusion between clients and server allows an **input-dependent abort attack** [Bar21]
- Without collusion, the Prover never learns the Verifier's output ("Accept", "Reject", "Abort")
- With collusion, the Prover can deviate to force an Abort as a function of the client's input

Threat Models:

- 2017 Kashefi-Pappa
  - Prepare-and-send with weakly quantum clients
  - Dishonest clients colluding with server left as an open problem
- 2021 Bartusek
  - Computational security
  - Malicious Blindness against collusion

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Our Approach

Our Goal:

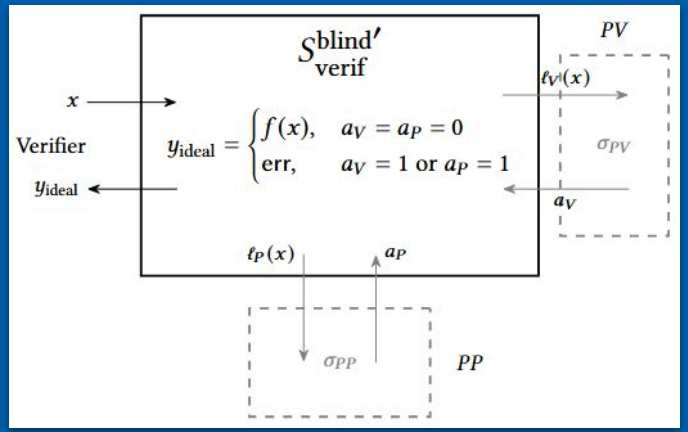
- We want multi-client functionality with best available multiple prover efficiency & security

Our Contribution:

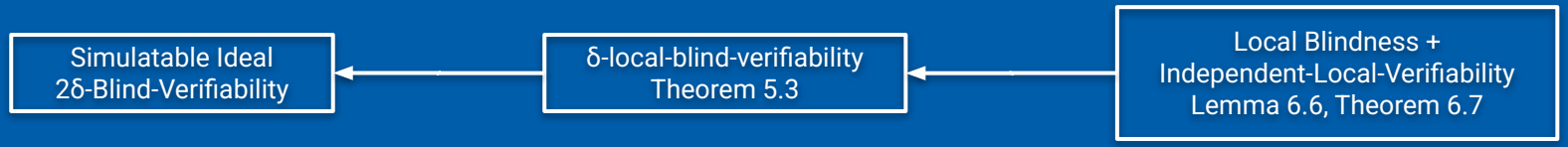
- Used a multi-prover base protocol and adapted it to be multiparty allowing collusion but still being resistant to an input-dependent abort attack
- Information-theoretic delegation security
- Used the AC framework to prove composability

Our Strategy:

- Modify the CGJV Leash protocol
- Ensure completeness and soundness are maintained
- Prove DFPR local properties
- Only changing V's internal behaviour



AC DFPR Reduction



Local Blindness:

- Prover's final state can be reproduced by a map acting on its own input

Local Verifiability:

- For any adversary P, the final state for V is either the correct state or an orthogonal error state

Independent Local Verifiability:

- Local verifiability + Prover's view and the abort flag are jointly simulatable

Adaptation to 2 Provers & Classical Verifier:

- Classical channels as quantum channels + dephasing
- 1 Honest prover restriction

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CGJV Verifier-on-a-Leash

Summary:

- Classical Verifier **V**, 2 entangled non-communicating Provers **PV** & **PP**
- Rigidity Branch + Delegation (Broadbent) Branch
- Circuit compiled into Clifford + T gates
- Simulatable blindness and Fault tolerance left as open problems

Self-testing/Rigidity Theorem:

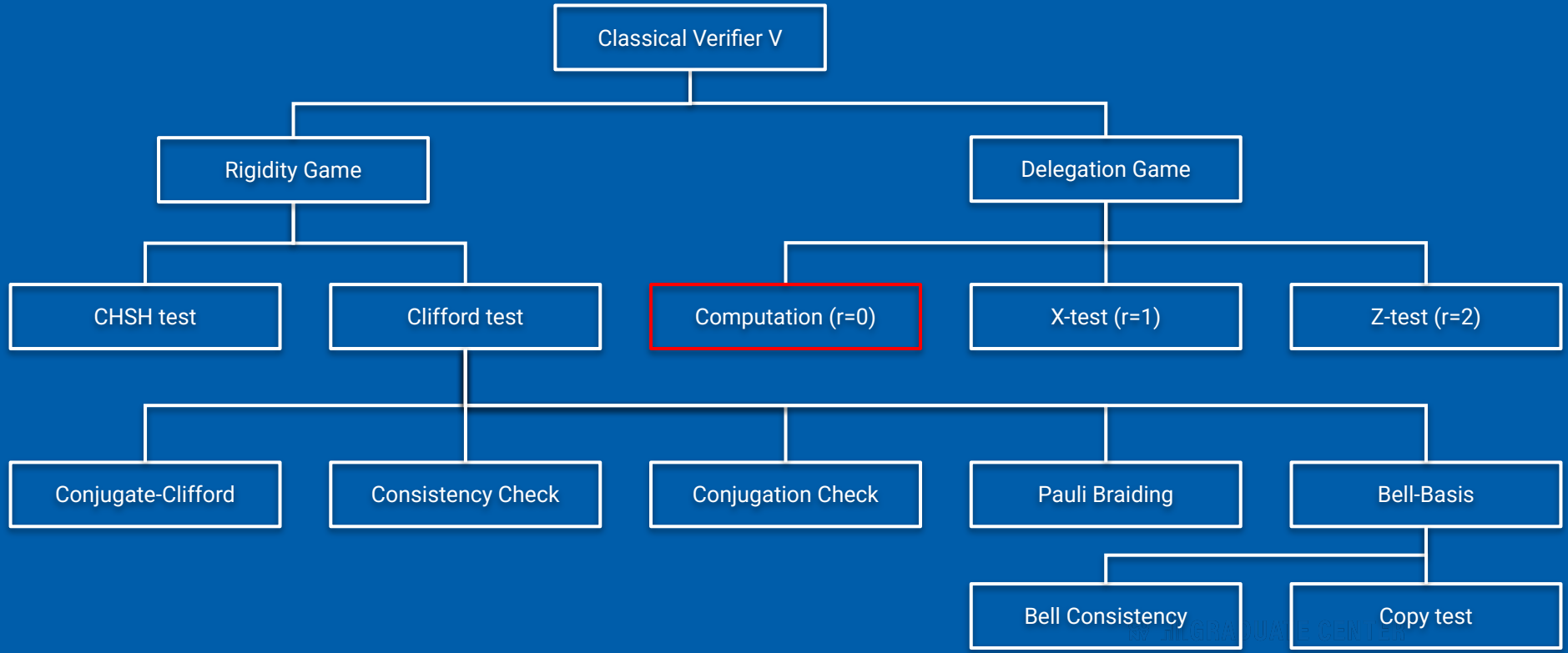
- Certifies tensor products of single qubit Clifford Observables with robustness independent of the number of qubits

Broadbent Structure:

- Computation Round: Server executes target circuit on arbitrary input
- X-test Round: Server executes identity circuit on |0>
- Z-test Round: Server executes identity circuit on |+>

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CGJV Verifier-on-a-Leash



CGJV Verifier-on-a-Leash

Sequential Version

- In CGJV, every branch outputs either “accept” or “reject” even the non-computational branch
- A sequential wrapper with Q and Q’ is used and statistics are tracked to map pairs of “accept-reject” and “reject-accept” to “YES” and “NO” instances respectively
- Other combinations are mapped to “Abort”
- “Abort” outcome is not guaranteed to be input-independent, no issue here as V does not broadcast outcome to PV or PP

We want “Abort” outcome to be input-independent

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Modifying CGJV

Extend the Output Alphabet:

- Rigidity branch & tests outputs now either err, or ok

Get single round Broadbent tradeoff:

$$\Pr[Y = 1 - f(x) | \text{comp}] \leq \frac{1}{3} + \frac{4}{3}\Pr[Y = \text{err} | \text{test}].$$

Change Sequential Wrapper:

- Introduce single round outcome counters
- p_r is the rigidity branch probability
- K is the number of repetitions
- R_H is the honest rigidity error
- R_0 is the rigidity error cutoff ($R_0 \geq 64\, R_H$)

$$M_b = \sum_{i=1}^{K} \mathbf{1}[y_i = b], \quad b \in \{0, 1\},$$

$$M_c = M_0 + M_1,$$

$$E_{\text{test}} = \sum_{i=1}^{K} \mathbf{1}[y_i = \text{err},\ g_i = \text{del},\ r_i \in \{1, 2\}],$$

$$E_{\text{rigid}} = \sum_{i=1}^{K} \mathbf{1}[y_i = \text{err},\ g_i = \text{rig}].$$

$$W_L(K) = \begin{cases} \text{err}, & E_{\text{test}} > 0, \\ \text{err}, & E_{\text{rigid}} > p_r R_0 K/32, \\ 0, & M_0 > M_1, \\ 1, & M_1 > M_0, \\ b_L, & M_0 = M_1, \end{cases}$$

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Leash' Security

Completeness:

- With K repetitions and honest provers we can output incorrect bit if:
  - Incorrect computation round majority (or tie-break)
- With K repetitions and honest provers we can output an error if:
  - Honest rigidity error may exceed threshold (2nd term)

$$\Pr[W_L(K) = f(x) | \text{honest}] \geq 1 - \left(1 - \frac{1-p_r}{3} + \frac{1-p_r}{3} e^{-1/18}\right)^K - e^{-p_r^2 R_0^2 K/2048}$$

Soundness:

- Use rigidity theorem to bound arbitrary PV to be close to the honest behaviour in Broadbent
- Arbitrary strategy must induce wrong computational majority without triggering rigidity or test errors
- Take conditional expectation values for incorrect outcome in computation round, error in test round, and error in rigidity round and bound the difference in expectation values

$$\Pr[W_L(K) = 1 - f(x)] \leq e^{-(1-p_r)^2 K/512\gamma^2} \qquad \lambda = \frac{1-p_r}{3R_0 p_r}, \quad \gamma = 1/2 + \max\{1, \lambda\}.$$

Leash' Security

AC Properties:

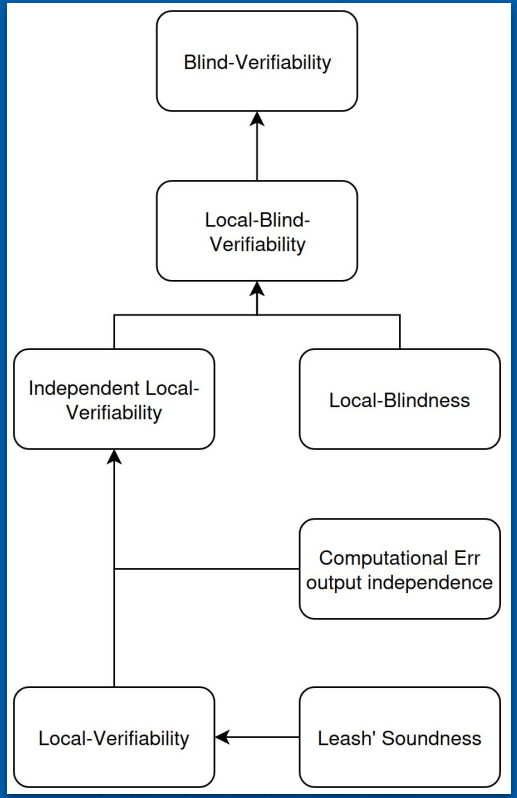
- Correctness uses the Completeness bound
- Simulatable Blind-Verifiability comes from local properties using the DFPR reduction
- Local-Verifiability comes from the Soundness bound

Honest PP:

- Local-Blindness
- Independent Local-Verifiability
  - Soundness + Dummy variable argument

Honest PV:

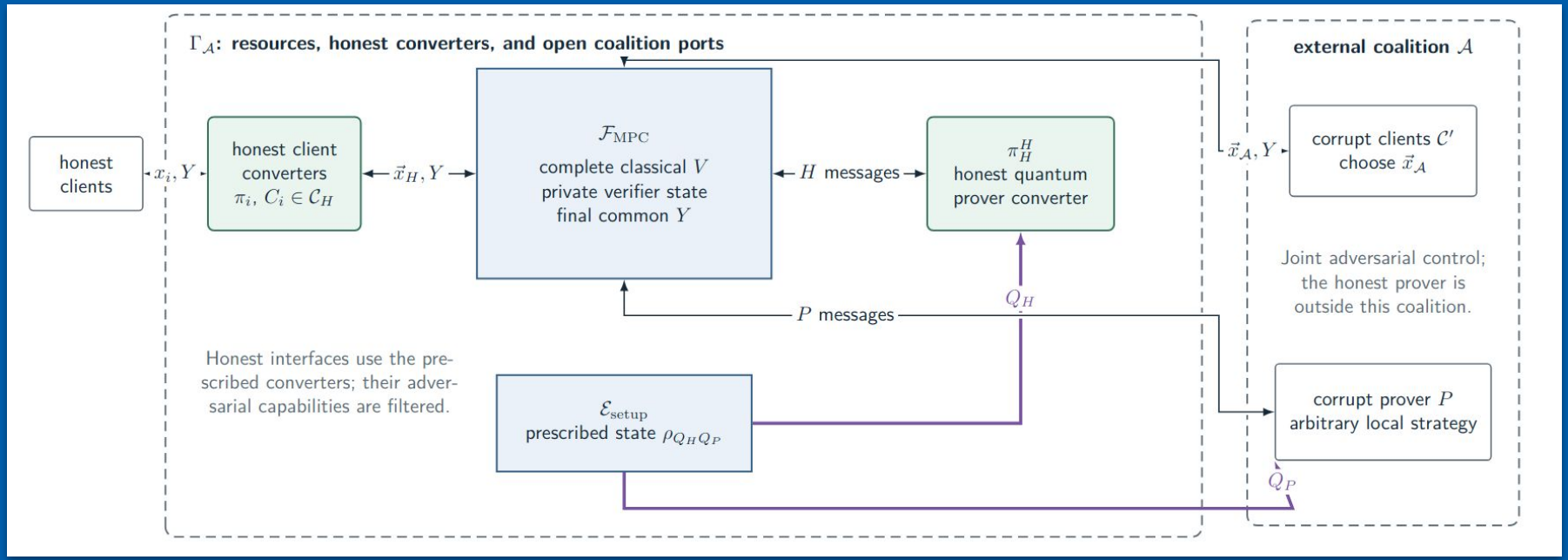
- Local-Blindness
  - From reordering argument
- Independent Local-Verifiability
  - Reordering + PP argument



Multi-client Composition

Based on the Kashefi-Pappa Approach

- Ideal MPC V functionality $\mathcal{F}_{MPC}$ ≈ Real MPC μ
- Hybrid Protocol Γ ≈ Ideal multiparty resource $S_{f,M,t}$ + simulator σ
- Composition gives: Real Protocol Γ^{μ} ≈ Ideal multiparty resource $S_{f,M,t}$ + simulator σ'



Future Directions

Conclusion:

- We have a multi-client VBDQC protocol with fully classical clients that is secure against client-server collusion

Open Problems:

- Remove 1 honest prover assumption - Global non-communicating distinguisher
- Optimize numerical parameters
- Derive explicit client functionality and real MPC
- Adapt AC security approach to new more efficient computationally secure protocols

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